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Review Articles

Systematic review: Effectiveness of combination of lactic acid attractants for control of dengue vector *Aedes spp.*

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ABSTRACT

Background & objectives: This study aimed to review the effectiveness of lactic acid when combined with other types of attractants for *Aedes spp.*

Methods: A systematic review was conducted according to the protocol for a systematic review and meta-analysis (PRISMA). Literature search used Cinahl, Medline/PubMed, ScienceDirect, ProQuest and Ebsco electronic databases. Research articles used in the systematic review were experimental articles that reported the effectiveness of mosquito traps using lactic acid or a combination of lactic acid with other attractants.

Results: From a total of 42 articles reviewed, there were 6 articles fulfilling the inclusion criteria. The highest synergistic combination of lactic acid in the ketone group was shown in the acetone compound, in the sulfides class, dimethyl sulfides, and in the chloroalkanes group, chloroform. The combination of lactic acid with two effective attractants can be seen in the incorporation of ammonia + caproic acid, and for the incorporation of lactic acid with three other effective attractants illustrated by combining ammonia + caproic acid + CO₂.

Interpretation & conclusion: Lactic acid as an attractant can be combined with other various compounds (ketone compounds, sulfides and chloroalkanes). Lactic acid increases its effectiveness in trapping *Ae. aegypti* and/or *Ae. albopictus* if combined with acetone, dimethyl sulfides, and/or chloroform.

Key words *Aedes spp.*; Attractant; Lactic Acid

BACKGROUND

Dengue fever has spread worldwide and affects millions of people every year in tropical and subtropical regions of Africa, Asia, Europe and America. Since there is no current effective vaccine against the dengue virus, prevention of disease transmission depends entirely on regulating the vector (*Aedes aegypti*) or interrupting human-vector contact¹. Therefore, development of new methods that are efficient, rational, effective and acceptable to the community is needed to reduce the number of people with DHF. Combining lactic acid with various other types of attractants mosquito traps can be an alternative to lure *Aedes aegypti* mosquitoes. Research conducted in the last decade for mosquito attractants has mainly studied the use of carbon dioxide (CO₂). This attractant is preferred because it is simple and relatively inexpensive. However, in subsequent studies the incorporation of CO₂ and lactic acid gave significant results, increasing the effectiveness of mosquito traps². Lactic acid is one compound for which attraction can be increased in the laboratory by combining

it with a variety of other types of attractants such as fatty acids³, ammonia, hexanoic acid, and octenol⁴, 1-octen-3-ol⁵, and CO₂⁶.

Ae. aegypti mosquitoes have evolved a remarkable innate preference for human odor that helps them locate and bite their prey. Various components of human skin extract, breath and sweat such as L-lactic acid⁷, ketones including acetone, butanone, and cyclopentanone², and small chain aliphatic acids⁸ have been shown to attract anthropophilic mosquitoes such as *Ae. aegypti*. Lactic acid has been shown to be a sign of human presence compared to other mammals^{7,9}. Ammonia has been proved to attract *Ae. aegypti* in combination with lactic acid⁴. A number of synthetic mixtures containing lactic acid, ammonia, short carboxylic acid, and ketones/sulfides/chloroalkanes have been reported to attract female *Ae. aegypti*². Furthermore, the synergism between a mixture of binary lactic acid and acetone or butanone² and a mixture of lactic acid, ammonia and ketones¹⁰ were also reported as luring agents for female *Ae. aegypti*.

The use of lactic acid mixed with the most effective

types of attractants to attract mosquitoes is still controversial and there are only a few studies comparing mixing of lactic acid with other types of attractants. This study aimed to review the effectiveness of lactic acid when combined with other types of attractants for *Aedes* spp.

MATERIAL & METHODS

The systematic review method began with a thorough search to determine whether other systematic reviews have been published on topics related to the effectiveness of lactic acid when combined with other types of attractants for *Aedes* spp. mosquito traps. Systematic reviews start from a list of definitions, search algorithms, abstract screening forms and data characterization and utility forms (DCU). Search locations for articles were originating from 5 databases, viz. Cinahl, Medline / PubMed, Science Direct, ProQuest and Ebsco for all publishing years using the keywords: trap or trapping mosquitoes or mosquitoes and lactic acid. Systematic search was done on all titles, abstracts and conclusions, and also on letters, editorials, and reviewing articles and any interactions with experts in their fields. The review team consisted of various multidisciplinary skills.

This systematic review aimed to answer the question whether the combination of lactic acid attractant is effective for the control of dengue fever vector *Aedes* spp. Keywords were implemented in a bibliographic database on January 23 2019. Article searches were limited to publications in English. The PubMed / Medline database obtained 20 articles. In the ProQuest and Science direct databases, 152 and 112 articles were obtained, respectively. And, in the Ebsco and Cinahl database there were 63 articles and 286, respectively. So, the total number of articles was 633 articles. After identifying 633 articles, a review of the titles of the articles was conducted. 53 articles were identified as duplicate. Articles that have potentially relevant titles were then reviewed by the abstract, while irrelevant articles were excluded. A total of 538 articles were excluded based on the title and abstract review. Articles with relevant abstracts were then reviewed in full-text. A total of 42 articles were reviewed in full-text. From this review, only 6 articles were screened for the systematic review and 36 articles were excluded. Of the 36 articles, 25 articles were excluded because the vectors were not *Aedes* spp., and 11 articles because their outcomes were not related to lactic acid (Fig. 1).

The screening steps were done by filtering keywords, titles and abstracts. The primary peer-reviewed article was considered relevant if the article discussed one or more aspects of the research question. Inclusion criteria were:

English-language articles, full text articles and original articles related to the study aim. Exclusion criteria were: duplicate article, the output was not lactic acid and the vector was not *Aedes* spp.

Full articles from citations that are potentially relevant were reviewed using the JBI form *Critical Appraisal Checklist for Quasi Experimental Studies* (Non randomized experimental studies) which consists of 9 potential questions to confirm the relevance of articles, data utility and extraction of main characteristics including information reported in the form of clarity of research on variables: Is there a control group, is there a measurement of results before and after intervention/exposure, is the follow-up complete or not and adequately analyzed, are cases and controls measured in the same way, are the results of measurement reliable and whether statistical analysis is appropriate to use.

Data were extracted from each article selected and used to inform the effectiveness of lactic acid both as a single attractant or in combination with other types of attractants. The profile of lactic acid activity both as a single attractant or in combination with other types of attractants was then summarized. All potentially relevant citations identified by literature searches were imported into reference management software (RefWorks 2.0). Duplicate search results were deleted manually. Data collected on the DCU form were exported to Microsoft Excel spreadsheets, formatted, and analyzed descriptively (frequency and percentage) to facilitate categorization, charting, and discussion.

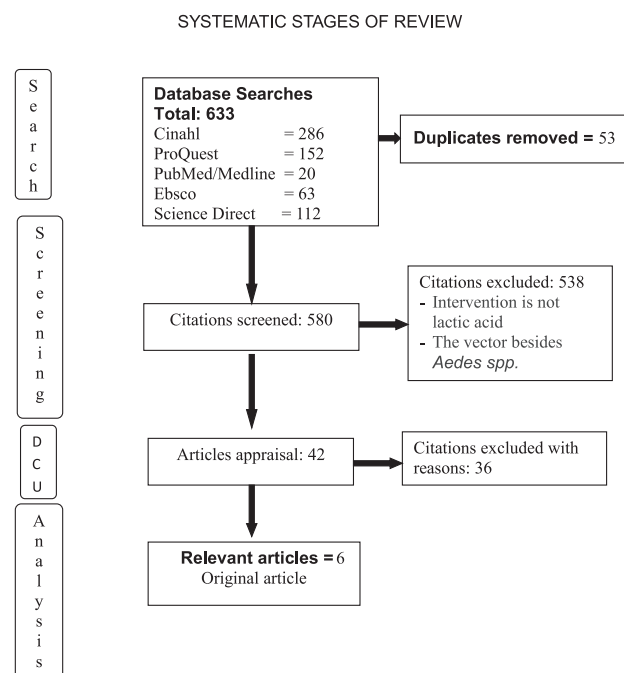


Fig. 1: Study Selection Process

Table 1. The combination of lactic acid in various types of attractants for Dengue vector *Aedes* spp.

No	References	Country	Purpose	Treatment	Mixture mean $\pm$ SE (%)	Single mean $\pm$ SE (%)	Summed mean $\pm$ SE (%)	Species	Information
1	12	USA	Laboratory Comparison of <i>Aedes aegypti</i> Attraction To Human Odors and To Synthetic Human Odor Compounds And Blends	ACE+DMDS	84.9 $\pm$ 2.6	0.2 $\pm$ 0.2	14.9 $\pm$ 2.1	<i>Ae. aegypti</i>	Synergist
				DMDS	69.4 $\pm$ 6.9ab	0.4 $\pm$ 0.3	30.2 $\pm$ 7.0	<i>Ae. aegypti</i>	Synergist
2	14	USA	Evaluation of Carbon dioxide, 1-Octen-3-OL, and Lactic acid as baits in Mosquito magnet™ Pro traps for <i>Aedes albopictus</i> in North Central Florida	OC	45.6 $\pm$ 10.8a	17.8 $\pm$ 4.4b	-	<i>Ae. albopictus</i>	Synergist
				LA	-	27.8 $\pm$ 7.5ab	-	<i>Ae. albopictus</i>	-
3	13	USA	Semi-field comparison of the BG lure, nonanal, and 1-octen-3-ol to attract adult mosquitoes In North-western Florida	BG Lure	202.0 $\pm$ 49.0a	-	-	<i>Ae. albopictus</i>	Synergist
				BG Lure + CO ₂	208.0 $\pm$ 48.6a	-	-	<i>Ae. albopictus</i>	Synergist
				BG Lure+ nonanal	21.3 $\pm$ 6.1c	-	-	<i>Ae. albopictus</i>	Antagonist
				BG Lure+ nonanal + CO ₂	123.3 $\pm$ 16.9a	-	-	<i>Ae. albopictus</i>	Antagonist
				BG Lure+ Octenol	63.5 $\pm$ 8.7b	-	-	<i>Ae. albopictus</i>	Antagonist
				BG Lure+ Octenol + CO ₂	286.5 $\pm$ 58.3a	-	-	<i>Ae. albopictus</i>	Synergist
4	11	China	Preliminary analysis of several attractants and spatial repellents for the mosquito, <i>Aedes albopictus</i> using an olfactometer	DCM	44.6 $\pm$ 2.8a	25.5 $\pm$ 2.6b	-	<i>Ae. albopictus</i>	Synergist
				ACE	40.5 $\pm$ 1.6a	15.3 $\pm$ 2.8c	-	<i>Ae. albopictus</i>	Synergist
				LA	-	10.1 $\pm$ 3.5c	-	<i>Ae. albopictus</i>	-
				Human odor	-	31.2 $\pm$ 2.1b	-	<i>Ae. albopictus</i>	-
5	4	USA	Comparison of Trapping performance between the original BG-Sentinel trap and BG-Sentinel 2 Trap	LA + ammonia+hexanoic acid	105	-	-	<i>Ae. albopictus</i>	Synergist
				LA + ammonia + hexanoic-acid + OC	40	-	-	<i>Ae. albopictus</i>	Antagonist
				OC	-	16	-	-	-
6	2	USA	Laboratory studies of <i>Ae. aegypti</i> attraction to ketones, sulfides, and primary chloroalkanes tested alone and in combination with L-lactic acid.	ACE	77.1 $\pm$ 4.2a	33.7 $\pm$ 4.6*	56.2 $\pm$ 2.5*	<i>Ae. aegypti</i>	Synergist
				Butanone	68.7 $\pm$ 5.8b	24.7 $\pm$ 3.2*	47.2 $\pm$ 2.1*	<i>Ae. aegypti</i>	Synergist
				3-Pentanone	36.7 $\pm$ 4.5c	10.5 $\pm$ 2.0*	33.0 $\pm$ 3.0	<i>Ae. aegypti</i>	Synergist

Ketones Group

2-Pentanone	32.7 ± 4.4cd	9.6 ± 2.5*	32.1 ± 3.5	<i>Ae. aegypti</i>	Synergist
3-Hexanone	31.7 ± 3.2cd	9.8 ± 2.1*	32.3 ± 3.1	<i>Ae. aegypti</i>	Synergist
2-Tridecanone	25.0 ± 3.3cde	7.0 ± 1.6*	29.5 ± 3.1	<i>Ae. aegypti</i>	Synergist
2-Hexanone	23.4 ± 3.7def	8.7 ± 1.6*	31.2 ± 3.2	<i>Ae. aegypti</i>	Synergist
5-Nonanone	21.6 ± 3.2defg	13.1 ± 3.4	35.7 ± 2.7*	<i>Ae. aegypti</i>	Antagonist
4-heptanone	21.5 ± 3.7def	12.2 ± 1.7*	34.7 ± 3.2*	<i>Ae. aegypti</i>	Antagonist
2-heptanone	20.0 ± 3.2defgh	12.3 ± 2.0*	34.8 ± 3.2*	<i>Ae. aegypti</i>	Antagonist
3-oktanone	17.9 ± 2.8efgh	19.2 ± 3.9	41.7 ± 3.1	<i>Ae. aegypti</i>	Antagonist
3-heptanone	17.2 ± 3.5efgh	11.5 ± 1.8	34.0 ± 3.1*	<i>Ae. aegypti</i>	Antagonist
2-dedacanone	14.5 ± 2.7efgh	7.5 ± 1.9*	30.0 ± 3.5*	<i>Ae. aegypti</i>	Antagonist
2-octanone	13.3 ± 1.7efgh	12.4 ± 2.8	34.9 ± 2.5*	<i>Ae. aegypti</i>	Antagonist
3-decanone	10.4 ± 1.3fgh	10.9 ± 2.3	33.4 ± 2.9*	<i>Ae. aegypti</i>	Antagonist
4-decanone	9.1 ± 1.5fgh	7.2 ± 0.9	29.8 ± 3.6*	<i>Ae. aegypti</i>	Antagonist
2-nonanone	7.7 ± 2.0gh	8.6 ± 1.8	31.1 ± 3.1*	<i>Ae. aegypti</i>	Antagonist
3-nonanone	7.5 ± 1.7gh	7.5 ± 1.6	30.0 ± 3.2*	<i>Ae. aegypti</i>	Antagonist
2-undecanone	7.4 ± 2.2gh	2.9 ± 0.9	25.5 ± 3.8*	<i>Ae. aegypti</i>	Antagonist
2-decanone	6.4 ± 0.9h	10.7 ± 1.5*	33.2 ± 3.4*	<i>Ae. aegypti</i>	Antagonist
L-lactic acid	-	22.5 ± 4.4def	-	<i>Ae. aegypti</i>	-
DMS	80.0 ± 3.2a	44.8 ± 5.2*	58.9 ± 3.2*	<i>Ae. aegypti</i>	Synergist
DMDS	70.5 ± 2.6ab	31.8 ± 3.7*	45.9 ± 2.0*	<i>Ae. aegypti</i>	Synergist
AMS	67.6 ± 8.8ab	15.1 ± 4.8*	29.1 ± 3.0	<i>Ae. aegypti</i>	Synergist
Carbon disulfide	62.8 ± 9.1ab	13.9 ± 5.2*	28.0 ± 3.9*	<i>Ae. aegypti</i>	Synergist
DMTS	50.2 ± 9.7bc	27.5 ± 6.4*	41.6 ± 2.4	<i>Ae. aegypti</i>	Synergist
Methyl propyl disulfide	37.1 ± 8.2cde	21.1 ± 3.1	35.2 ± 3.5	<i>Ae. aegypti</i>	Synergist
Ethyl disulfide	26.6 ± 4.5cde	12.1 ± 2.0*	26.2 ± 4.1	<i>Ae. aegypti</i>	Synergist
Allyl disulfide	22.6 ± 8.8de	10.9 ± 6.3	25.0 ± 5.1	<i>Ae. aegypti</i>	Synergist
LA	-	14.1 ± 5.7e	-	<i>Ae. aegypti</i>	-
Chloroalkanes Group					
Chloroform	81.1 ± 5.7a	45.6 ± 4.2*	59.7 ± 3.4*	<i>Ae. aegypti</i>	Synergist
1.1.2-trichloroethane	80.1 ± 3.2a	26.7 ± 7.7*	40.7 ± 4.9*	<i>Ae. aegypti</i>	Synergist
Dichloromethane	76.5 ± 5.7a	41.4 ± 5.9*	55.5 ± 2.5*	<i>Ae. aegypti</i>	Synergist
Carbon tetrachloride	72.7 ± 8.4a	28.1 ± 5.9*	42.2 ± 2.8*	<i>Ae. aegypti</i>	Synergist
Trichloroethylene	71.5 ± 4.7a	25.8 ± 6.0*	39.9 ± 2.3*	<i>Ae. aegypti</i>	Synergist
Carbon dioxide	68.1 ± 5.1a	8.4 ± 3.6*	22.5 ± 3.7*	<i>Ae. aegypti</i>	Synergist
Tetrachloroethylene	62.5 ± 9.5ab	11.4 ± 2.9*	25.4 ± 3.8*	<i>Ae. aegypti</i>	Synergist
1.1.1-trichloroethane	59.6 ± 4.5ab	13.3 ± 4.9*	27.4 ± 4.3*	<i>Ae. aegypti</i>	Synergist
LA	-	14.1 ± 5.7	-	<i>Ae. aegypti</i>	-

LA, L-lactic acid; ACE, acetone; DMDS, dimethyl disulfide; DMS, Dimethyl sulfide; DMTS, Dimethyl trisulfide; OC, Octenol; BG Lure (Ammonia, Caproic acid + Lactic acid); DCM, Dichloromethane.

RESULTS

On June 10 2019, 42 articles were identified that reported the effectiveness of lactic acid to lure mosquitoes and finally 6 articles were filtered according to the systematic feasibility of the review. These six selected studies were published between 2007 and 2015. The year of publication, location and combination of lactic acid and other attractants are described in table. There was one study conducted in Asia¹¹ and five studies from the USA^{2, 4, 12–14}. In general, all the research included in this review are studies that aimed to describe the level of effectiveness of lactic acid combined with other types of attractants.

Literature survey showed that lactic acid can be combined with dozens of other attractant compounds such as the ketone group, sulfides and chloroalkanes group-sin attracting *Aedes* spp (Table 1). Several compounds in ketone group (Acetone, Butanone, 3-Pentanone, 2-Pentanone, 3-Hexanone, 2- Tridecanone and 2-Hexanone) were found to be synergistic when combined with lactic acid in luring *Ae. aegypti* while other compounds in ketone groups were antagonists to lactic acid. All compounds in sulfide and chloroalkane groups that have been tested and combined with lactic acid had synergistic tendencies to attract *Ae. aegypti*. Other studies for *Aedes* spp. collection indicated *Ae. albopictus* was the most prevalent species of the traps fed with lactic acid and octenol (45.6 ± 10.8) and the lowest collection came from traps which were only fed octenol (17.8 ± 4.4). Medium collections were obtained in traps which were only fed with lactic acid (27.8 ± 7.5)¹⁴ (Table 1). *Ae. albopictus* collection demonstrated the significant importance for the treatment of octenol + lactic acid. The addition of lactic acid to octenol feed with CO₂ increases the collection of *Ae. albopictus*. Based on these results the reduction of the population of *Ae. albopictus* by using traps fed with octenol by adding lactic acid needs to be studied more deeply. The majority of species was *Ae. albopictus* collected in traps by combining the attractant octenol + lactic acid or a trap that only uses lactic acid to lure the mosquitoes. The use of octenol as an attractant is related to the decreasing of the collection of *Ae. albopictus*, while the addition of lactic acid feed to octenol traps has a synergistic effect on the trap collection (Table 2).

The combined chemical test of L-lactic acid, acetone, and dichloromethane had an average capture rate of 10.1%, 15.3%, and 25.5%, respectively (Table 1)¹¹. On the other hand, L-lactic acid and acetone were less effective in attracting adult *Ae. albopictus* with a range of 10–15% lower compared to the expression by the scent of the human body¹¹. The combination of L-lactic acid and dichloromethane, and L-lactic acid and acetone are far more

Table 2. Collection of *Aedes albopictus* in Mosquito Magnet Pro Traps producing carbon dioxide and baited with lures producing lactic acid or octenol

Mean ($\pm$ SE) ¹ Collected/ trap period		
Treatment	<i>p</i>	Explanation
LA	0.06	related
OC	0.33	No reaction
OC + LA	0.03	Synergist
Control	0.12	No reaction

¹Means compared by ANOVA ($p < 0.05$).

LA, L-lactic acid; OC, Octenol.

effective in the attraction of adult female *Ae. albopictus* rather than the three chemicals by themselves¹¹. A mixture of 1% L-lactic acid in acetone yields a capture rate of 40.2, and is comparable to a mixture of 1% L-lactic acid in dichloromethane, which results in a capture rate of 43.8%¹¹. Further increase in L-lactic acid up to 10% in acetone and dichloromethane did not produce a significant increase in the number of catches of *Aedes* spp. mosquitoes¹¹.

Another study of the use of attractants showed that the average total of mosquitoes captured per trap / day was the highest for traps with bait 3C (Lactic acid + ammonia + hexanoic acid) followed by bait 4C (Lactic acid + ammonia + hexanoic acid + octenol), and then traps with feed 1C (octenol) (Table 1)⁴. This pattern informs that the more attractants used do not have a positive correlation with the number of catches of *Aedes* spp. However, another research that combined 3 attractants (ammonia + caproic acid + CO₂) demonstrated more synergistic effects when compared with the combination of 2 attractants (ammonia + caproic acid)¹³. The inhibition of *Aedes* spp. host searching ability can be due to high doses, so that there may be irritation in mosquitoes when detecting the host's odor¹⁵. Thus, the combination of several attractants in attracting *Aedes* spp. needs to be studied more.

Research with a combination of acetone and L-lactic acid (Table 1) demonstrated the most alluring binary mixture and the mixture with butanone is a little less alluring². Ketone attractant compounds when together with L-lactic acid show synergism between the two, while the percentage of attractiveness is greater than the number of their single treatment. A decrease in the attractiveness of the mixture is shown in C5 and C6 ketones, and surprisingly, the next highest level of attraction is in the mixture with C13, 2-tridecanone ketones. For ketones in the range C7 to C12, there is attraction that is decreased from the mixture, indicating that this combination causes resistance to attraction in *Ae. aegypti* mosquitoes¹². Table 1 contains data from examination of sulfides, disulfides, and dimethyl trisulfide. Both monosulfides tested, dimethyl sulfide, and

allyl methyl sulfide produce a synergistic mixture to lure *Ae. aegypti* mosquitoes. Carbon disulfide and dimethyl disulfide also produce a synergistic mixture. However, other disulfides and dimethyl trisulfides are not mixtures that produce a level of attraction that indicates synergism for the *Ae. aegypti*. Another aspect was that there were significant chemical differences between internal trisulfide, disulfide, monosulfide and carbon disulfide. Table 1 illustrates a high level of attraction to chloride, which means it acts at a synergistic level when given together with L-lactic acid. For all of these chlorides, the combination with L-lactic acid significantly increases the attraction, and all mixtures are synergistic at the dose level tested.

The combination of lactic acid in the ketones group with the highest level of effectiveness in attracting species of *A. aegypti* was shown in the compound acetone, while in the sulfides class, namely dimethyl sulfides, and in the chloroalkanes group, namely chloroform. As for the incorporation of lactic acid with two effective attractants it is seen in the incorporation of ammonia + caproic acid, and for the incorporation of lactic acid with three other effective attractants it was illustrated by the incorporation of ammonia + caproic acid + CO₂. There is some evidence that other factors can influence the attraction of *Ae. aegypti* to get close or stay away from these attractants.

DISCUSSION

Mosquito attractants are useful in managing mosquito populations so as to reduce the spread of vector-borne diseases. The use of attractants in mosquito traps is very promising^{16–17}. However, the effectiveness of attractants must be carefully evaluated before practical application. Different mosquito species or even different geographical sub-populations of certain species, can react differently to the chemicals given^{11,18–19}. At present, the testing of potential attractants that can be applied to mosquito traps has been focused on single compounds including on the *Ae. aegypti* mosquito. Humans produce hundreds of chemical compounds from various classes of compounds^{20–21}. Some of them produce lower levels of attraction in bioassays and greater attractiveness when mixed into binary mixtures and triners, or triple compound mixtures¹².

Lactic acid is especially attractive to *Aedes* spp.^{10,22–23}, because they have lactic acid sensitive receptors on their antenna²⁴. Additionally, another study using a mixture of lactic acid and octenol feed together with CO₂ in the trap of Mosquito Magnet Pro results in a higher collection than in trapping with octenol or lactic acid alone¹⁴. The use of lactic acid + octenol with CO₂ in the Pro Mosquito Magnet trap produces more *Ae. albopictus* than that collected

from octenol-only traps. Further study of mosquito attraction using lactic acid traps will be useful in determining their impact on the mosquito population.

Three attractants were used to analyze the effectiveness of each of the three attractants, L-lactic acid, acetone, and dichloromethane, and compared to each other. The three chemicals are effective in attracting *Aedes* spp. females. However, the effectiveness of these three chemicals varies greatly. For example, dichloromethane is most effective, resulting in capture rates comparable to human odors. On the other hand, L-lactic acid, the main metabolite of human skin, only produces a 10% catch rate, the lowest among all three chemicals. The effectiveness of the combination of these three chemicals is very different from the use of a single chemical. When 1% lactic acid is added to acetone or dichloromethane, the catch rate increases to more than 40%, higher than human odor¹¹. This finding suggests that gradient proportions play an important role in the effectiveness of attractants in attracting mosquitoes to mosquito traps. Another aspect to note is that the olfactometer system, optimized test conditions, and the combination of chemical attractants chosen will help the effectiveness of mosquito traps in the future.

In another finding it was revealed that inhibition of host search could be due to high doses so that there is some possibility of irritation to mosquitoes when detecting the attractant odor¹⁵. The presence of naturally occurring repellent effects of metabolic byproducts on human skin was reported by Khater *et al.*,²⁵ and described more by other researchers^{2,26}. These findings suggest that skin lipids or some carboxylic acids, aldehydes, and heterocyclics inhibit the ability to search for hosts by disrupting the ability of insects to find attractive sources of odor²⁷.

Addition of lactic acid to both ketones (acetone and butanone) resulted in a synergistic effect¹⁰. The study showed that a mixture of lactic acid, acetone, and butanone is the most promising attractive mixture of all other combinations. Trinary mixtures of lactic acid, acetone, and butanone are significantly preferred over their respective binary mixtures. In the case of comparisons between mixtures containing acetylacetone, mixtures containing acetone or butanone as one of the components are preferred over those containing acetyl acetone¹⁰. These results highlight the fact that a complex blend is more preferred than its individual components by *Ae. aegypti* for host seeking.

CONCLUSION

Lactic acid as an attractant of *Aedes* spp. can be combined synergistically with various other compounds (ke-

tone compounds, sulfides and chloroalkanes). Lactic acid increases its effectiveness in attracting *Ae. aegypti* and/or *Ae. albopictus* when combined with acetone or dimethyl sulfides and/or chloroform compounds.

Conflict of interest: None

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